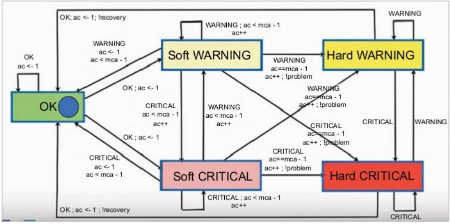

Figure 1: State diagram

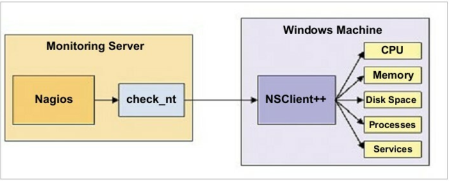

Figure 2: NSClient++ communicating with NagiosXI

as CPU and memory. The monitoring can be carried out at the host and network level to report system statistics. The intelligent platform management interface (IPMI) stack is used for hardware monitoring. Host monitoring can be carried out using Nagios, Ganglia or other related software. HPC cluster monitoring integrates the IPMI stack with Nagios. The unified framework summarises the hardware monitoring of the server to the Nagios Web interface.

The IPMI sub-system functions autonomously, irrespective of the OS, and permits administrators to monitor the system remotely without the presence of the OS or the management application. The system can be monitored continuously and it abstracts the data from local area networks when the power source is connected. The IPMI prescribes only the structure and format of the interfaces as a standard, while the deployment varies.

Monitoring checks

The monitoring checks can be categorised into local and remote. Local checks include the following aspects:

- Monitoring the Windows desktop
- Monitoring Windows server

Remote checks include the following aspects:

- Monitoring website URLs
- Google Map integration with NagiosXI

Monitoring the Windows desktop

On a Windows desktop, NagiosXI monitors the following:

- Memory usage
- CPU load
- Disk usage
- Service states
- Running processes

Monitoring private services or attributes of a Windows machine requires the installation of an agent. The agent acts as a proxy between the NagiosXI plugins, which do the monitoring of the actual service or attribute of the Windows machine. We have used NSClient++ as the add-on on the Windows machine and are using the *check_nt plugin* to communicate with the NSClient++ add-on.

How NSClient++ communicates with NagiosXI is shown in Figure 2.

How to configure NSClient++ in Windows: After downloading the NSClient++ 4.0.2, we have to install it and, during installation, specify the IP address of our NagiosXI machine as well as the NSClient password.

Configuring NSClient in NagiosXI: After installing NSClient++ in Windows, we have to make some changes in the NagiosXI configuration wizard and select the Windows desktop option, before setting the IP address of the Windows desktop in NagiosXI.

Settings for the desktop metrics: We can set our own parameters for monitoring the desktop. For CPU, memory usage and disk usage, we have to set our own threshold value, below which the 'soft state' will be converted into the 'hard critical' and 'hard warning' state. The desktop metrics settings are shown in Figure 3.

Desktop monitoring settings: We can determine the maximum number of checks for the desktop. The NagiosXI will keep on checking till the count reaches the maximum count value, after which NagioXI will notify the admin about the error with the help of alert messages and e-mails.

Settings for desktop notifications: Once an error has been detected, the admin should be informed on time before the end user gets to know. Various methods are used for notifying the admin. We can create our own method of notification using e-mails, SMS alerts, etc.

After configuring all the initial settings, we can show the status of CPU usage, disk drives, etc. The checks display the states of various monitored parameters. They also display the maximum checks for each device.